

Technical Indicators & Strategy Backtesting with Alpaca: AAPL

Strategy Descriptions and Rules

Buy & Hold: Invests in the selected ticker throughout the full sample period.

Trend Following: Buys when MACD is above its signal line, ADX is greater than 25, and SMA20 is above SMA50. Sells when MACD falls below signal or SMA20 falls below SMA50.

Mean Reversion: Buys when RSI is below 30 and price is below the lower Bollinger Band. Sells when RSI is above 70 or price is above the upper Bollinger Band.

Custom Strategy: Combines trend, momentum, and volume. Buys when price is above SMA50, MACD is above signal, RSI is between 45 and 70, CMF is positive, and OBV is above OBV SMA20. Sells on trend weakness or extreme RSI conditions.

Performance Comparison

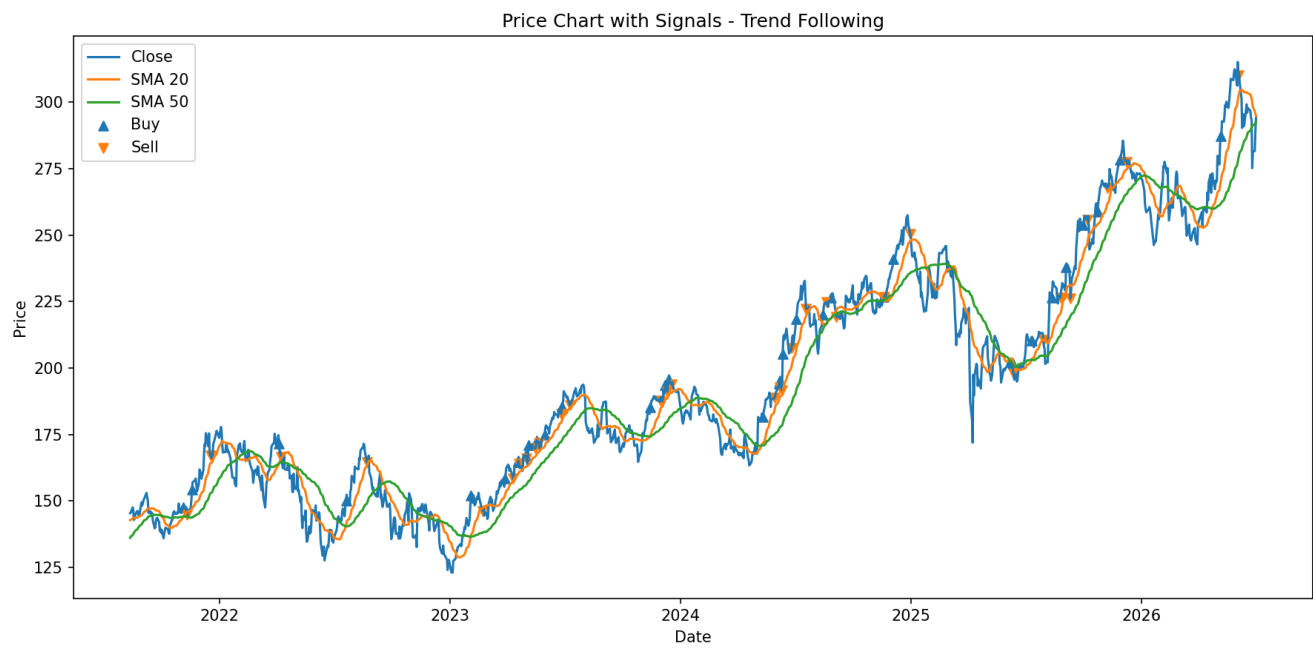
Strategy	Total Return	CAGR	Volatility	Sharpe Ratio	Sortino Ratio	Max Drawdown	Win Rate	Trades
Buy & Hold	102.57%	15.62%	27.75%	0.66	0.96	-33.25%	N/A	1
Trend Following	23.73%	4.47%	9.12%	0.53	0.39	-9.78%	50.00%	32
Mean Reversion	92.10%	14.36%	18.00%	0.83	0.62	-24.07%	85.71%	15
Custom Strategy	14.06%	2.74%	9.73%	0.33	0.23	-23.56%	43.75%	48

Discussion of Results

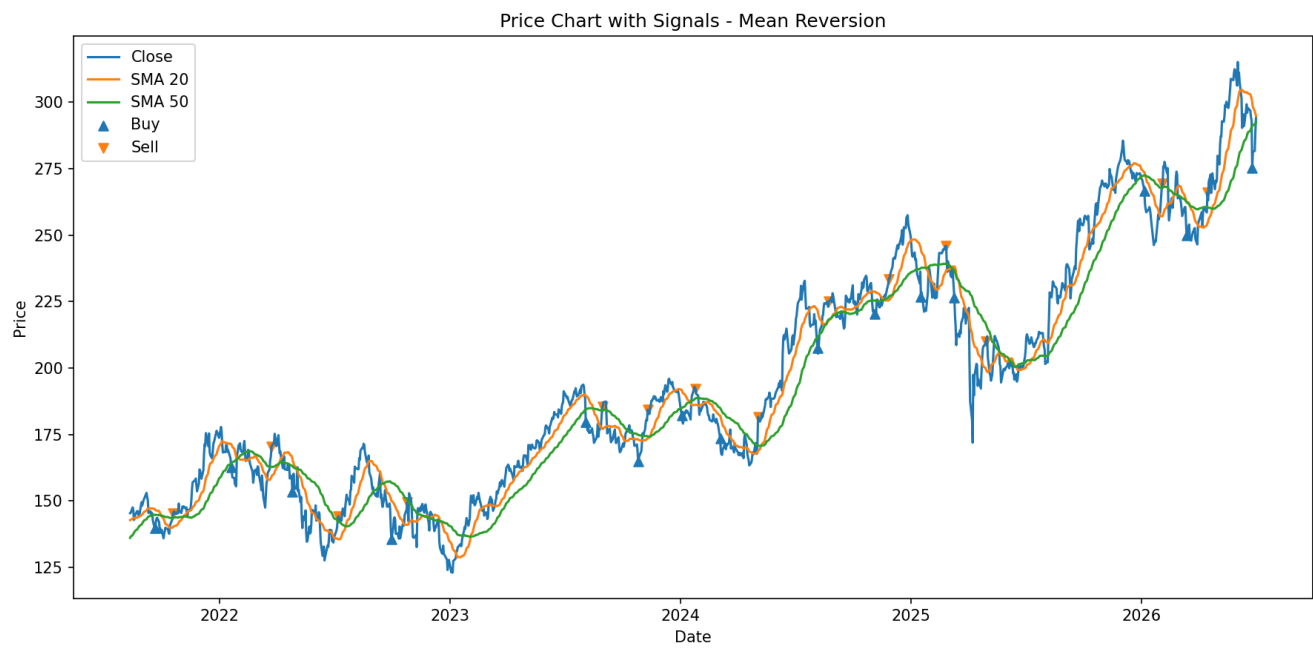
Based on Sharpe Ratio, the best risk-adjusted strategy in this backtest is Mean Reversion with a Sharpe Ratio of 0.83. The strategy with the highest Sortino Ratio is Buy & Hold with a Sortino Ratio of 0.96. For the AAPL backtest, Mean Reversion is selected as the best active trading strategy because it has the highest Sharpe Ratio among all tested strategies and produced strong total return with fewer trades than the trend-following and custom strategies. Buy & Hold had the highest total return and Sortino Ratio, but it also experienced the largest maximum drawdown. This suggests that Mean Reversion offered the strongest overall risk-adjusted active strategy profile for this ticker and sample period. The final conclusion depends on the selected ticker and sample period. Because the engine uses previous-day positions for next-day returns, the backtest avoids look-ahead bias. However, this simplified project does not include transaction costs, slippage, taxes, or survivorship-bias controls.

Charts

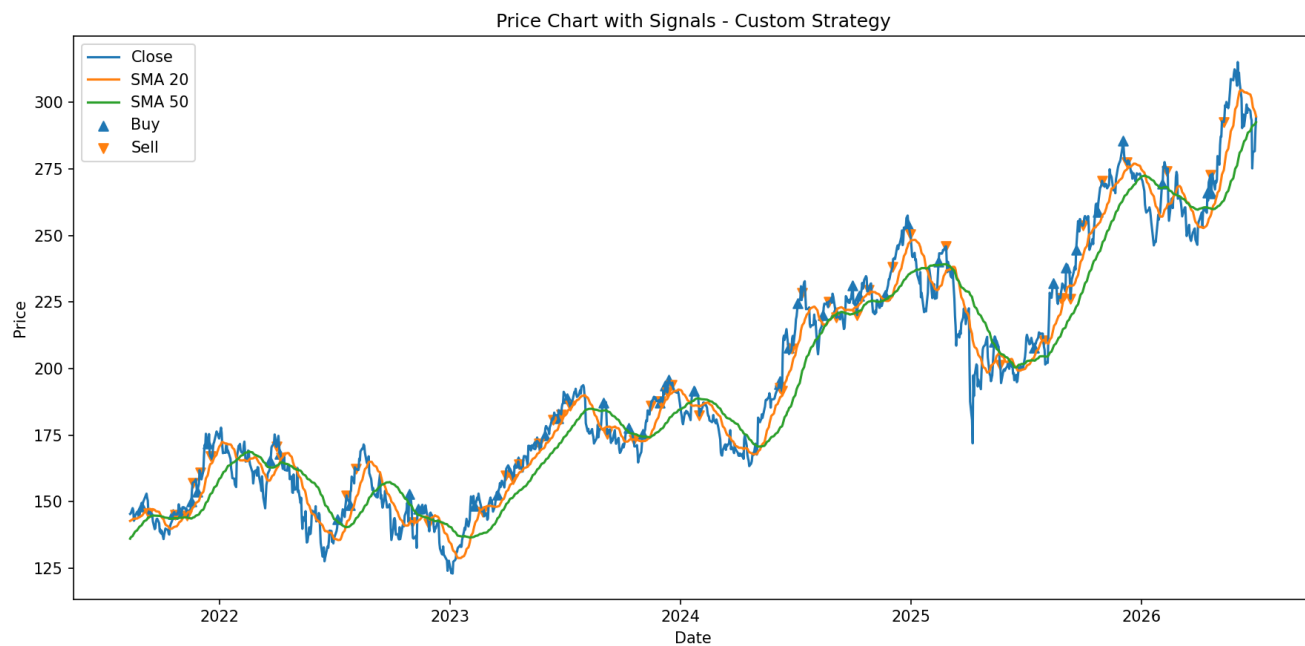
price_signals_trend_following.png



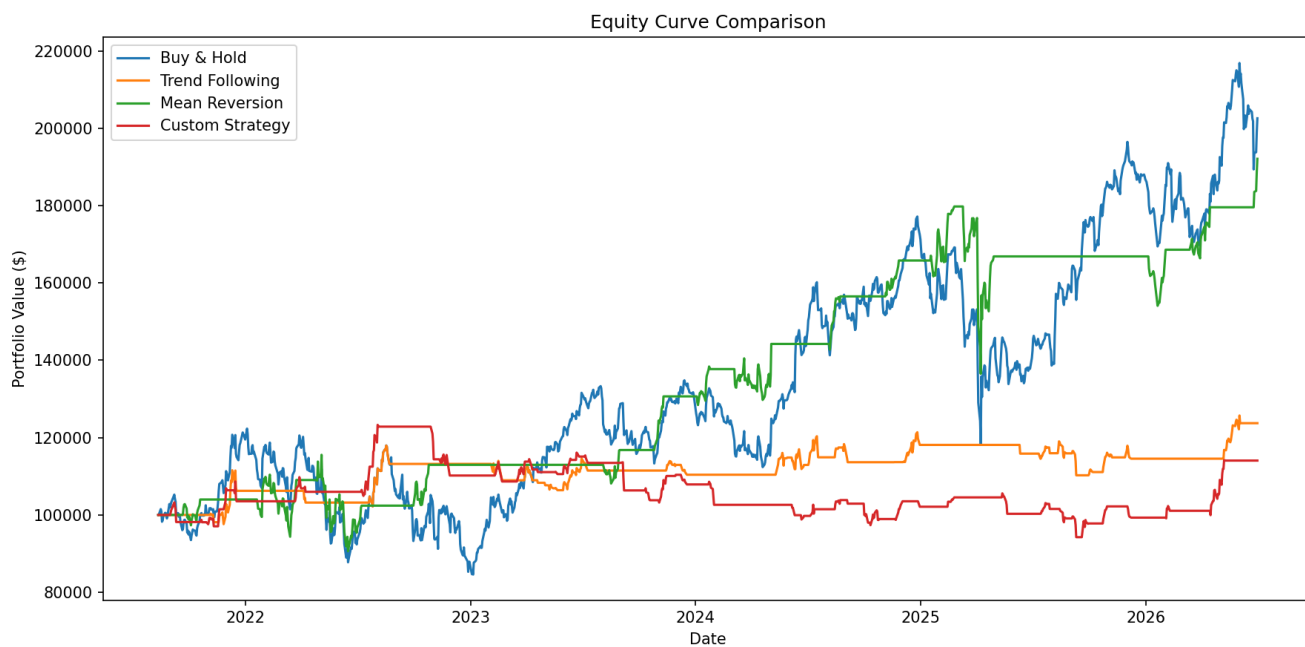
price_signals_mean_reversion.png



price_signals_custom_strategy.png



equity_curves.png



drawdowns.png

